1 Vectors in Two and Three Dimensions

Question: What information specifies a line?

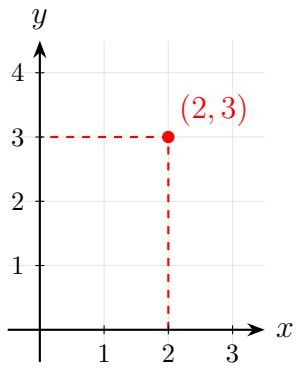
- 2 points
- A point and a direction vector
- Slope and y-intercept
- Intersection of 2 non-parallel planes

Comparison:

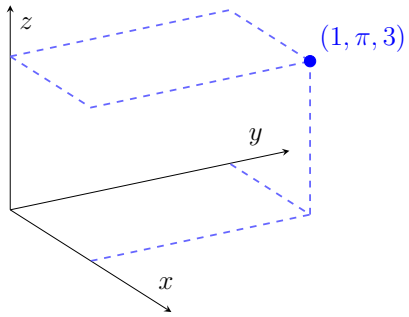
- **Single-variable calculus:** $f(x) \to$ rate of change of x
- **Multivariable calculus:** $f(x, y) \to$ functions of multiple variables

1.1 Coordinates

2-dimension (2D): $(2, 3)$



3-dimension (3D): $(1, \pi, 3)$



1.2 Vectors and Position Vectors

Definition 1.1: Equivalent Vectors

Two vectors are **equivalent** if one is a translation (pure shift without rotation) of the other.

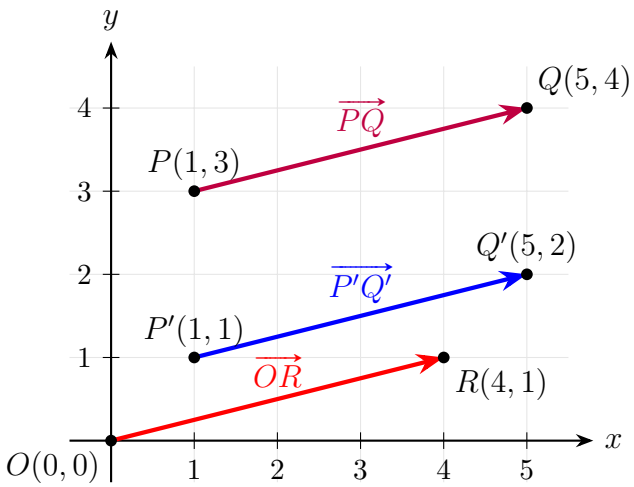
Example 1.1.

Given points $P = (1, 3)$, $Q = (5, 4)$ and $P' = (1, 1)$, $Q' = (5, 2)$. Let $R = (4, 1)$ and

$O = (0, 0)$. Then:

$$\overrightarrow{PQ} = \overrightarrow{P'Q'} = \overrightarrow{OR}$$

This could be indicated by the diagram below:



Definition 1.2: Position Vector

A vector whose initial point is at the origin $O(0, 0)$ is called a **position vector**. It is written using angle brackets:

$$\overrightarrow{OR} = \langle 4, 1 \rangle$$

Definition 1.3: Magnitude

The **length** or **magnitude** of a vector $\vec{v}$, denoted by $\|\vec{v}\|$, is the length of the line segment between its endpoints.

If $\vec{v} = \langle v_1, v_2 \rangle$, then its magnitude is given by:

$$\|\vec{v}\| = \sqrt{v_1^2 + v_2^2}$$

For example, using $\overrightarrow{PQ} = \langle 4, 1 \rangle$:

$$\left\|\overrightarrow{PQ}\right\| = \sqrt{4^2 + 1^2} = \sqrt{17}$$

1.3 Unit Vectors

Definition 1.4: Unit Vector

A vector $\vec{v}$ is called a **unit vector** if its magnitude is 1, i.e., $\|\vec{v}\| = 1$.

In general, a unit vector of any function is given by:

$$\hat{v} = \frac{\vec{v}}{\|\vec{v}\|}$$

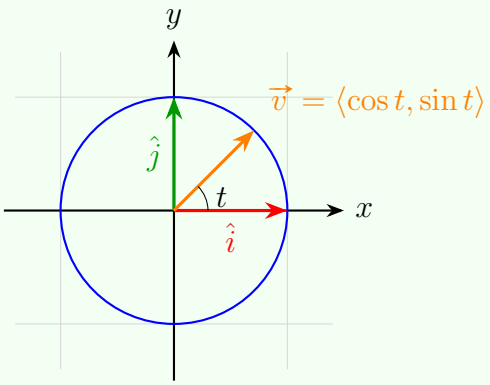
Problem 1.1: Describing Unit Vectors

Describe all unit vectors in $\mathbb{R}^2$.

Solution 1.1

A vector $\vec{v} = \langle x, y \rangle$ is a unit vector if and only if $x^2 + y^2 = 1$. Equivalently, using polar parameterization:

$$\vec{v}(t) = \langle \cos t, \sin t \rangle \quad \text{for } 0 \leq t < 2\pi$$



Standard Basis Vectors:

- **2D Basis Vectors:** $\hat{i} = \langle 1, 0 \rangle, \quad \hat{j} = \langle 0, 1 \rangle$
- **3D Basis Vectors:** $\hat{i} = \langle 1, 0, 0 \rangle, \quad \hat{j} = \langle 0, 1, 0 \rangle, \quad \hat{k} = \langle 0, 0, 1 \rangle$

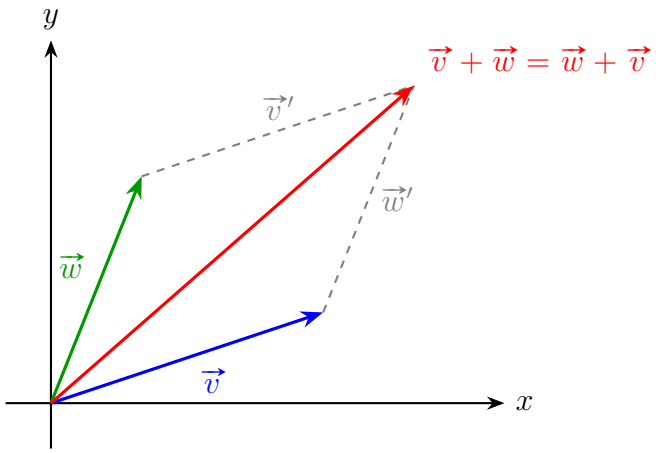
1.4 Vector Operations

Addition and Subtraction:

$$\langle v_1, v_2\rangle + \langle w_1, w_2\rangle = \langle v_1 + w_1, v_2 + w_2\rangle$$
$$\langle v_1, v_2\rangle - \langle w_1, w_2\rangle = \langle v_1 - w_1, v_2 - w_2\rangle \quad \text{(Equivalent to } \vec{v} + (-1)\vec{w}\text{)}$$

Scalar Multiplication: For any scalar $\lambda \in \mathbb{R}$ and vector $\vec{v} = \langle v_1, v_2\rangle$:

$$\lambda\vec{v} = \langle \lambda v_1, \lambda v_2\rangle$$



Important Note

Vectors are **not** designed to be divided! (Vector division is undefined).

1.5 Parallel Vectors

Definition 1.5: Parallel Vectors

Two non-zero vectors $\vec{v}$ and $\vec{w}$ ($\vec{v}, \vec{w} \neq \langle 0, 0\rangle$) are *parallel* (denoted as $\vec{v} \parallel \vec{w}$) if there exists a non-zero scalar $\lambda \in \mathbb{R}$ such that:

$$\vec{w} = \lambda\vec{v}$$

Equivalently, we have a geometric definition.

Definition 1.6: Parallel Vectors (Geometric)

$\vec{v} \parallel \vec{w}$ if and only if both vectors define or lie on the *same line passing through the origin.*